

the surface of the object to be viewed. Information on temperature difference, available in the form of IR energy, is collected by the thermal imaging device and converted into an electronic image. The ability to detect tiny temperature differences that exist not only between the desired object and the surroundings, but also between different points of the object itself, coupled with emission in IR region allows a thermal imaging device to see in near total darkness.

Thermal imaging night vision devices are extensively used not only by military and law enforcement agencies for target detection and acquisition, surveillance and monitoring, search and rescue operations, fire-fighting and so on, these are also used in fields such as medicine, archaeology, process monitoring, automotive industry, meteorology and astronomy.

Image intensifier tube. An image intensifier tube amplifies low-light images to levels that can be seen with the human eye or detected by a digital image sensor. An image intensifier tube collects the existing ambient light originating from natural sources such as starlight or moonlight, or from artificial sources such as streetlights or IR illuminators, through the objective lens of the night vision device.

Low-level light consisting of photons enters the night vision device through the input window and strikes the photocathode. Inside of the input window is generally coated with a thin layer of light-sensitive material. This layer acts as the photocathode, which is protected from any damage from oxidation by operating the image intensifier tube under vacuum of the order of 10^{-9} to 10^{-10} torr. Photo electrons released by the photocathode are accelerated and focused by a high magnitude electric field towards the microchannel plate (MCP). The MCP has millions of small channels and the electrons entering these channels are both accelerated by another high magnitude electric field within the MCP and multiplied by secondary emission resulting from electrons bouncing off the inner walls of these channels.

For each electron entering the MCP, approximately 1000 electrons are generated and, subsequently, accelerated from the output of the MCP by a third

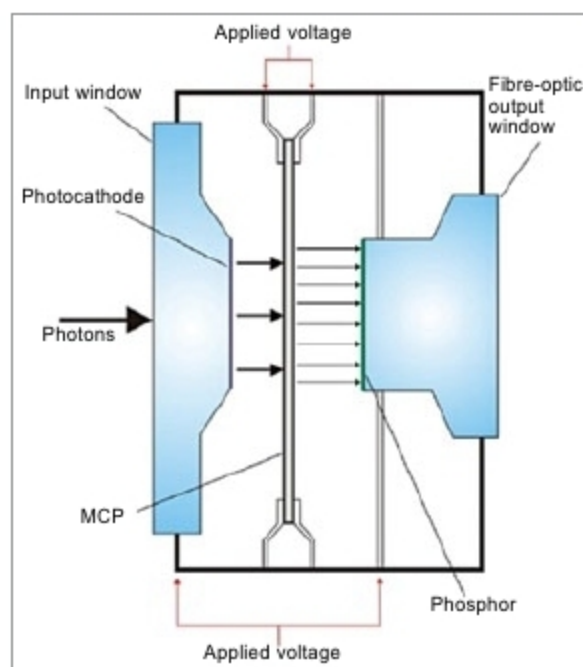


Fig. 1: Construction of an image intensifier tube

electrical field towards the phosphor screen. The phosphor screen, which is a thin light-emitting layer deposited on the inside of the output window of the image intensifier tube, converts the impinging electrons back to photons. For every photon entering the input window of the intensifier tube, tens of thousands of photons come out of the output window after emission from the phosphor screen.

This photon multiplication takes place due to electron acceleration in the region between the input window and the photocathode, electron acceleration and secondary emission within the MCP channels, and electron acceleration in the region between the MCP and the phosphor screen. This multi-stage process produces an intensified or amplified image of the object that is much brighter than the original image.

Construction. Constituent parts of an image intensifier tube include input window, photocathode, MCP, phosphor screen, output window and power supply. Fig. 1 gives an overview of the constructional features of an image intensifier tube.

Input window material is selected according to required sensitivity at shorter wavelengths. Common materials used for input window are synthetic silica (transmitting wavelength of 160nm or longer), fibre-optic plate (with transmission wavelength of 350nm or longer), magnesium fluoride (with transmission wavelength of 115nm or longer) and borosilicate glass (with transmission wavelength of 300nm or longer).

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